

WEEK 06 · REFERENCE

THE RULES, IN ONE PLACE

Thresholds, decision trees and the three scales. Keep this one; it answers the questions you will actually have in week 10.

WHAT IS IN HERE

Every threshold this week names, the four states with their failure modes, and the three scales.

HOW TO READ IT

First column is the case you are in. The last one is what it costs, or what to do.

WHEN TO OPEN IT

Mid-build, when a state, a focus stop or a contrast ratio has become a guess.

► **THE WORKING TOOL IS ELSEWHERE:** `audit-checklist-he.md` — forty-five rows, printable. This deck is the reasoning behind them.

CONTRAST: THREE NUMBERS, AND WHAT EACH COVERS

RATIO	APPLIES TO	WHERE PEOPLE GET IT WRONG
4.5 : 1	Normal text, up to 23px	Placeholder text and disabled labels are still text
3 : 1	Large text — 24px, or 19px bold	Assuming a big heading is exempt. It is not; it is 3:1
3 : 1	Non-text: the edge of a control, the focus ring, a meaningful icon	Almost everybody. A page can pass on every word and have fields nobody can find
—	Decorative lines, disabled controls, logos	Over-applying it: a 3:1 divider under every card reads as a 1990s table

Measure against the colour actually **painted** behind the text, which is often an ancestor's background rather than the element's own.



THE FOUR STATES, AND HOW EACH ONE FAILS

STATE	OWES THE USER	THE USUAL FAILURE
Empty	What this is, and what to do next	A blank rectangle — or one panel for both empties , so “no results” shows before anything was searched
Loading	That something is happening, roughly how much, and a deadline	A spinner with no timeout: it spins until somebody reloads
Error	What happened · what to do · a way out	Error . Or a status code. Or silence
Success	The data, and where it came from	Local fallback data presented as though the network answered

In code this is **one field with four values**, not three booleans.

`isLoading + hasError + isEmpty` allows eight combinations, four of which are meaningless — and one of those four will happen.

FEEDBACK LATENCY: WHAT TO SHOW, AND WHEN

HOW LONG	WHAT THE USER NEEDS
under 100ms	Nothing. It feels instant; a spinner here is noise
100ms – 1s	The result. Do not animate a wait this short
1s – 10s	A skeleton or a progress indicator, and the control disabled
over 10s	Progress with an estimate, and a way to cancel
8s in this course	Give up and move to the error state. A request with no timeout can hang forever

Also: a control the user just activated must acknowledge **immediately**, even if the result takes a second. An unacknowledged click gets clicked again.

VALIDATION TIMING

EVENT	RIGHT FOR	WHY
<code>input</code>	Filtering a local list	Every keystroke shows more of what you wanted. No wrong intermediate state
<code>input</code>	Never for first validation	“Not a valid email” after two characters is true, useless, and trains people to ignore messages
<code>blur</code>	Validation , first time	They have finished. Now the message is about a state they chose to leave
<code>input</code> , after invalid	Validation, second time onward	So the message clears the moment it is fixed, not when they leave again
<code>submit</code>	Anything touching the network	Rate limits are real: one endpoint in the pool allows one request per second

DESTRUCTIVE ACTIONS

UNDO — THE DEFAULT

Costs nothing up front. Covers the deletion they meant and regretted. Needs a notice that survives long enough to be seen — and focus moved into it.

CONFIRM — THE EXCEPTION

Only when the action **cannot be undone**. Use a real `<dialog>` with `showModal()`, so the browser traps focus and Escape closes it.

NEITHER — THE FAILURE

A delete button identical to a cancel button, with nothing between the click and the loss. This scores zero on the rubric row.

A confirmation charges all ninety-nine intentional deletions for the one that was not — and after a week it is clicked through unread.

THE KEYBOARD CONTRACT

KEY	MUST DO
Tab / Shift+Tab	Move to the next / previous control, in document order
Enter	Activate a link or a button. Submit a form from inside a field
Space	Activate a button. Toggle a checkbox. Not a link
Arrows	Move within a group: radios, a select, a tab list
Escape	Close whatever opened last, without committing anything
Home / End	First / last in a group, where a group exists

You get every row of this **for free** from `button`, `a[href]`, `input`, `select` and `fieldset`. Every row you have to write yourself is a row you can get wrong.

FOCUS MANAGEMENT

- On open** Move focus into the new thing — the dialog, the panel, the first field.
- While open** Keep it in there. `showModal()` does this for you; `show()` does not.
- On close** Put it back where it came from. The half everybody forgets.
- On destroy** Catch it. After a delete the focused element is gone and the browser drops focus to `body`.
- Never** Steal focus on page load, or on a timer. Both throw the user out of whatever they were doing.

WHERE TO SEND IT AFTER A DELETE

To whatever the user would most plausibly do next — which, after a removal, is **undo**. Not the top of the page.

POSITIVE TABINDEX

`tabindex="1"` pulls an element to the front of the whole document. Everything stays reachable and the order becomes nonsense. Use `0` or `-1`, never more.

DOES THIS NEED ARIA?

1 Is there a native element for it?

Then use it. `button`, `dialog`, `details`, `fieldset` + radios.

2 Is the name already visible text?

Then say nothing. `aria-label` would **replace** it, and the two users are now reading different applications.

Is it a relationship the DOM cannot express?

3 A message belonging to a field: `aria-describedby`. A region that announces changes: `aria-live`.

This is what ARIA is for.

4 Are you about to add a role to a div?

Stop. It changes the announcement and not the behaviour — no Enter, no Space, no tab stop.

The first rule of ARIA is not to use ARIA. The second is that a wrong role is worse than none: it makes a promise the element does not keep.



ALT TEXT, AS A DECISION

1 Does the image carry information the text does not?

Describe that information. Not the picture — the information.

2 Decorative, or a duplicate of adjacent text?

`alt=""` — empty, present, deliberate. A cover beside its own title contributes nothing, and a non-empty alt makes it read twice.

3 Is it a link or a button's only content?

The alt **is** the control's name. Describe the destination, not the glyph.

4 Is it a chart?

The numbers go in text near it. An alt cannot carry a data set.

A MISSING ALT IS NOT THE SAME AS AN EMPTY ONE

No `alt` attribute at all makes a screen reader read the **file name**. `alt=""` removes the image from the accessibility tree entirely, which is what you wanted.

THE THREE SCALES

TYPE

One ratio, six steps. **1.2 to 1.333** for a page with a dozen elements; wider only for a poster. Under 1.12 is one size with rounding errors.

SPACE

One step, multiplied.

`--spacing: 0.25rem` gives **4 · 8 · 12 · 16 · 24 · 32 · 48**. Nothing between the steps, ever.

COLOUR

One hue. Saturation and **lightness** carry the roles — and lightness alone carries the contrast.

A scale is not a restriction; it is a decision taken once so it is not taken forty times. And rhythm is a property of the **set**: nobody can see that 17px is wrong, and everybody can see that 13, 18, 17, 22 has no rhythm.

MEASURE, AND THE TRAP IN `ch`

CHARACTERS PER LINE	READS AS
under 45	Work. The eye returns so often that long words break the rag apart
45 – 80	Comfortable. Aim here for anything longer than two lines
over 80	Tiring. On the way back the eye starts landing on the wrong line

`1ch` is the advance width of the digit **zero**, which is wider than the average lowercase letter — so a column set to `62ch` holds noticeably more than 62 characters of prose. `max-width: 62ch` is still the right instrument; just do not read the number as a count.



TESTING IT ON A PERSON

- 1 One person is enough to find the big thing.**
Five find almost everything. You do not need a lab.
- 2 Give a task, not a tour.**
“Register for the Tuesday guitar class.” Then stop talking.
- 3 Do not help.**
The silence where they are stuck is the finding. Helping destroys the only data you came for.
- 4 Watch the hands, not the face.**
Where the cursor hovers before it clicks is where they expected the control to be.
- 5 Ask afterwards, never during.**
“What did you expect that button to do?” — and their answer outranks your intention.

Not examined, and worth an hour anyway: **test your project on one human before week 12.** Whoever is next to you in the lab counts.



THE FIVE MISCONCEPTIONS

IT PASSES THE CONTRAST CHECKER

So one row of forty-five is green. The checker looked at `text` — it has no opinion about the edge of a field, the focus ring, or whether colour is the only cue.

WHITE SPACE IS WASTED SPACE

It is the cheapest instrument you own. Grouping is a **ratio** between inside and between — and 8 and 16 read exactly like 16 and 32.

`aria-label` MAKES IT ACCESSIBLE

It **replaces** the visible name. Put one on a button that already says “Send” and the two users are reading different applications. ARIA changes the announcement, never the behaviour.

VALIDATE AS THEY TYPE

Fourteen errors while somebody types one valid address. Every one true, every one about a state they were always going to pass through. Filter on `input`; validate on `blur`.

A SPINNER IS THE LOADING STATE

It is half of it. The other half is **what happens when it does not end**. No timeout means the spinner is the final state.

BONUS: `outline: none` LOOKS CLEANER

It removes the only thing telling a keyboard user where they are. If the default is ugly, **replace** it — `:focus-visible` fires for the keyboard and not for a mouse click, which is what you actually wanted.

WHERE TO LOOK, IN ORDER

1 The checklist

`audit-checklist-he.md` — forty-five rows, printable, with how to check each one.

2 The demos

`demos/01` ... `08` . Every one measures what it claims; open them again while you build.

3 The brief

`project-brief-he.md` — forty-one requirements, each a criterion you can test.

4 MDN

for the accessible-name computation and for `dialog` . Both are better documented than they are known.

5 A person

ten minutes with somebody who has not seen your project beats an hour with any of the above.